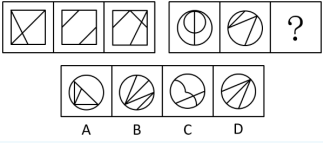


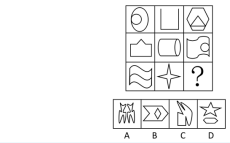
Quantitative Reasoning

From the four given options, select the most suitable one to fill in the question mark.



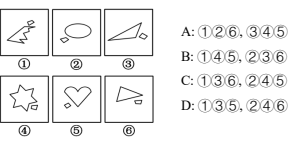
A B C D

From the given four options, choose the most suitable one to fill in the question mark.



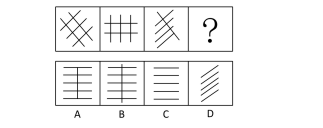
A B C D

Divide the following figures into two categories, ensuring that each category shares common patterns.



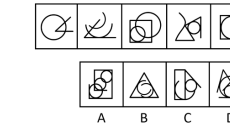
A: 126, 345
B: 145, 236
C: 136, 245
D: 135, 246

From the given four options, select the most suitable one to fill in the question mark.



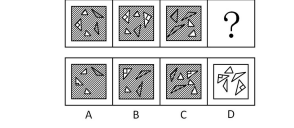
A B C D

From the given four options, select the most suitable one to fill in the question mark.



A B C D

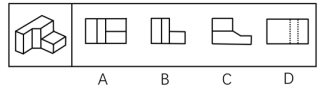
From the given four options, select the most appropriate one to fill in the question mark.



A B C D

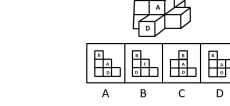
Spatial Reasoning

From any angle, which one on the right is not a view of the three-dimensional shape given on the left?



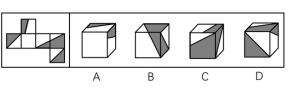
A B C D

Which of the following four options is the front elevation view of the figure in the question.



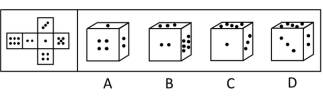
A B C D

The image on the left shows the net of a cube. Which of the options on the right can be formed by folding it?



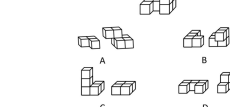
A B C D

The diagram on the left shows the net of a cube's outer surface. Please identify the incorrect option.



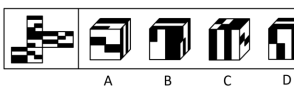
A B C D

Choose the option from the four given choices that cannot assemble into the 3D shape shown in the question.



A B C D

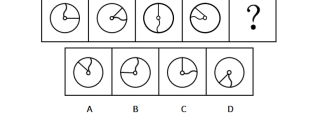
The left shows the net of a cube's outer surface. Which option on the right can be folded into it?



A B C D

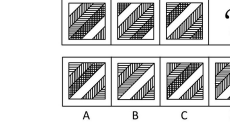
Positional Reasoning

From the four given options, select the most suitable one to fill in the question mark.



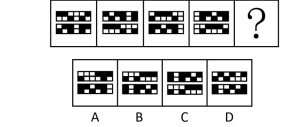
A B C D

From the four given options, choose the most suitable one to fill in the question mark to create a certain pattern.



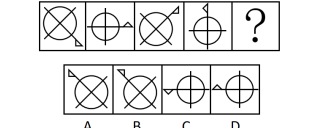
A B C D

From the four given options, choose the most appropriate one to fill in the question mark.



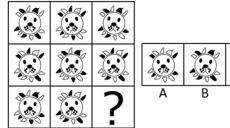
A B C D

The option that best matches the given pattern is ().



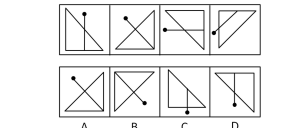
A B C D

From the given four options, choose the most suitable one to fill in the question mark.



A B C D

From the given four options, choose the most suitable one to present a certain regularity.



A B C D

Figure 11: More examples in VisuLogic of Quantitative Reasoning, Spatial Reasoning, Positional Reasoning.

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Attribute Reasoning

Divide the six figures below into two categories, so that each category of figures has its own common patterns.

① ② ③
④ ⑤ ⑥

A: ①②③, ④⑤⑥
B: ①②⑤, ③④⑥
C: ①③⑤, ②④⑥
D: ①④⑥, ②③⑤

From the four given options, choose the most suitable one to fill in the question mark.

A B C D

Divide the following figures into two categories, ensuring that each category shares common patterns.

① ② ③
④ ⑤ ⑥

A: ①②⑤, ③④⑥
B: ①③⑥, ②④⑤
C: ①②④, ③⑤⑥
D: ①③④, ②⑤⑥

Divide the six figures below into two categories, so that each category of figures has its own common patterns.

① ② ③
④ ⑤ ⑥

A: ①②③, ④⑤⑥
B: ①②⑥, ③④⑤
C: ①③④, ②⑤⑥
D: ①④⑤, ②③⑥

From the given four options, select the most suitable one to fill in the question mark.

A B C D

From the given four options, select the most appropriate one to fill in the question mark.

A B C D

Stylistic Reasoning

From the four given options, choose the most suitable one to fill in the question mark.

A B C D

Select the most suitable option from the four given choices to fill in the question mark.

A B C D

Choose the most suitable option from the four given choices to fill in the question mark.

A B C D

From the four given options, choose the most suitable one to fill in the question mark.

A B C D

From the four given options, choose the most suitable one to fill in the question mark.

A B C D

From the four given options, choose the most appropriate one to fill in the question mark.

A B C D

Other

From the four given options, choose the most suitable one to fill in the question mark: A: 2 B: 6 C: 4 D: 3.

From the given four options, choose the most appropriate one to fill in the question mark.

A B C D

Select the most suitable option from the four given choices to fill in the question mark.

A B C D

Choose the most suitable option from the four given choices to fill in the question mark.

A B C D

From the four given options, choose the most suitable one to fill in the question mark.

A B C D

From the four given options, choose the most appropriate one to fill in the question mark.

A B C D

Select the most suitable option from the four given choices to fill in the question mark.

A B C D

Figure 12: More examples in VisuLogic of Attribute Reasoning, Stylistic Reasoning, and Other.

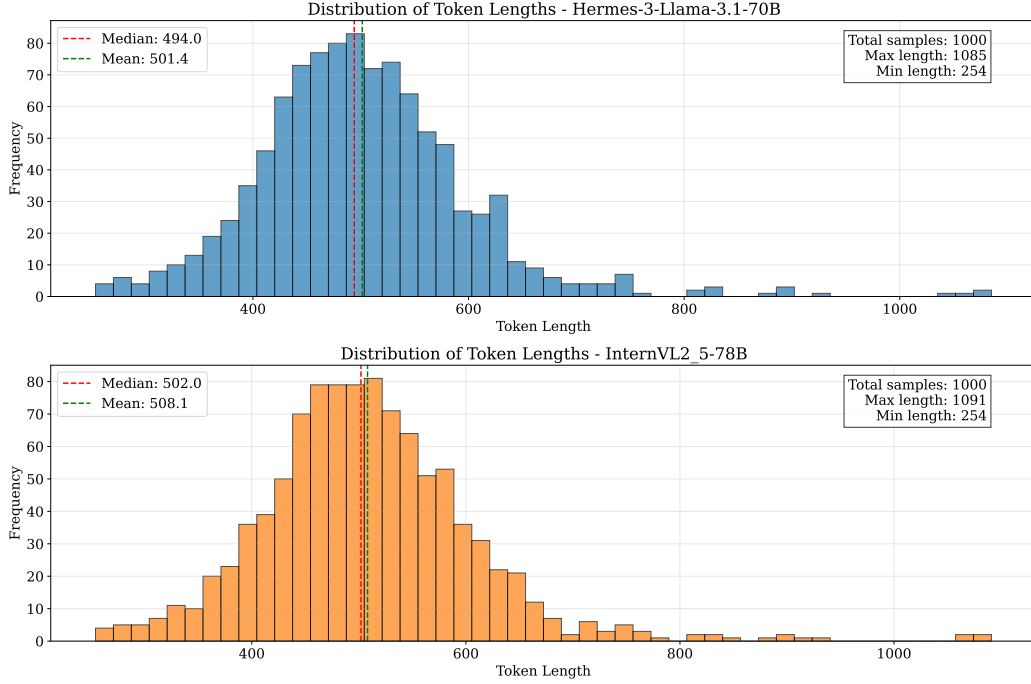


Figure 13: Distribution of tokens length in LLM evaluation settings, including image description.

Reward Modeling. Inspired by Deepseek-R1 [18], we design our rule-based reward system that mainly consists of two types of rewards:

1. **Format rewards:** To clarify model’s outputs, we design a format rule that forces model to put its thinking process between ‘<think>’ and ‘</think>’ tags and put its final answer between ‘<answer>’ and ‘</answer>’ tags. Regular expression is applied to judge whether outputs conform to the format rule.
2. **Accuracy rewards:** The accuracy reward is decided by the response’s correctness. The model should generate the response in right format, then the answer is extracted and judged whether it is matched to the correct option.

Hyperparameter settings. Our two RL models are trained with the hyperparameter configuration detailed in Table 5. And the hyperparameters used in SFT training stage are listed in Table 4.

Table 4: Hyperparameter Settings for SFT Training Stage.

	Qwen2.5-VL-7B-Instruct-SFT	InternVL2.5-38B-SFT
pretrain model	Qwen2.5-VL-7B-Instruct	InternVL2.5-38B
learning rate	0.5e-5	2e-5
batch size	64	128
optimizer	AdamW	AdamW
lr scheduler	cosine	cosine
image strategy	image_max_pixels=262144	max_dynamic_patch=6
warmup ratio	0.1	0.03
max epochs	1	1
bf16	True	True